

Final Project Proposal

Reddit as an Early Warning System: Detecting, Classifying, and Forecasting Real-World Events from Community Signals

DATS 6450 — Big Data Analytics | Spring 2026

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Problem Statement

Can we identify real-world events as they emerge by detecting anomalous surges in Reddit activity, sentiment shifts, and linguistic patterns—and classify their type, severity, and trajectory at scale? As a bonus exploration, can early precursor signals in community behavior forecast that a major event is about to materialize?

Dataset

Reddit comments and submissions from **June 2023 – July 2024** (~446 GB). The dataset spans thousands of subreddits, providing the volume and diversity needed for cross-community signal detection. Processing at this scale requires distributed computing—single-machine tools cannot handle rolling baselines across billions of records.

Technology Stack

Primary: Apache Spark on EC2 cluster (PySpark for EDA, Spark MLlib for ML, Spark NLP for text processing).

Supporting: AWS S3 for data storage, Python (matplotlib/plotly) for visualization, Quarto for the project website.

Research Questions

Exploratory Data Analysis

1. What does baseline Reddit activity look like per subreddit (hourly/daily volume), and how do we define statistically significant anomalies (e.g., z-score > 3 over rolling windows)?
2. How do anomalous volume spikes propagate across subreddits—do they originate in niche communities and spread outward, or emerge simultaneously across large subreddits?
3. What is the relationship between spike magnitude and post-spike engagement (score, reply depth, comment count), and can we build a taxonomy of spike shapes (impulse, plateau, slow build)?
4. Which time-of-day and day-of-week patterns correlate with different event types, and how do weekday vs. weekend spikes differ in propagation behavior?

Natural Language Processing

5. During detected anomaly windows, which named entities (people, organizations, locations) surge in frequency, and can entity co-occurrence graphs characterize event type?
6. How does subreddit-level sentiment shift during anomaly windows vs. rolling baseline, and does sentiment polarity distinguish event categories (disasters skew negative, launches skew mixed)?
7. Can topic modeling (LDA via Spark MLlib) on anomaly-window text automatically generate interpretable event labels or summaries?

Machine Learning

8. Given features (volume z-score, sentiment delta, entity concentration, cross-subreddit spread rate), can we classify detected events into categories: breaking news, controversy, product launch, meme/viral, disaster?
9. Can we predict whether a detected spike will sustain (>24h elevated activity) or decay rapidly, based on early-window features from the first 2–4 hours?
10. Bonus — Forecasting: Using precursor signals (gradual volume ramps, sentiment drift, entity emergence) in pre-event windows, can we predict whether a major event will materialize in the next N days? Framed as exploratory; strong results not expected for black-swan events.

Approach

We will manually curate 20–40 confirmed real-world events from the June 2023–July 2024 window as ground truth. The pipeline builds outward from EDA: rolling baselines and anomaly detection produce the features consumed by NLP (entity/sentiment extraction during anomaly windows) and ML (classification and trajectory prediction). The forecasting question (Q10) reuses the same feature set on pre-event windows and is scoped as exploratory—we will report precision/recall honestly and discuss which event types show forecastable precursor signals vs. those that do not.